

Formula project Roadmap

Design and build your own motor controller using linear vectorisation

- We have two motors which will be connected via direct drive to the rear wheels of the car - We would like to calibrate the turning and speed of the wheels to the steering wheel
- Instead of a mechanically powered steering system we would like to use electric control

Design a safety circuit that recognises if there is a flaw in the drivetrain

<https://www.imeche.org/docs/default-source/1-oscar/formula-student/2026/rules/fsuk-2026-rules---v1-09e21118e54216d0c8310ff0100d05193.pdf?sfvrsn=2>

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- This must follow Formula Student standards which includes logic based electronics
- The drivetrain must be safe and powerful enough to move a minimum of 200kg

Design a safety circuit that recognises if there is a flaw in the

Design a Shutdown Circuit

- If the drivetrain experiences a dangerous situation - the shutdown circuit should immediately cut power to the drivetrain

Project objectives / Directions

Motor Controller

Calibrate speed

Calculate turning

Calibrate power distribution from turning.

Safety Circuit

Finds flaws using gauges like volt meter, temperature gauge, incorrect figures

Safety Circuit shutdown

Once flaws are found shut down

What do I know?

It's rear wheel drive with electronically controlled steering and calibration.

Uses gauges to track internal systems, when internal systems fail shutdown.

Drive chain attached to the two back wheels independently, 2 motors one for each back wheel and 2 batteries.

Batteries connect to the motor controller then to the motors.

Goal

Their goal, movement

My goal, refined movement and on track.

Requiring body to build, my motor controller, drive chain and wheel set up

Hydrogen Fuel Cell Vehicle Drivetrain Project

Direction & Engineering Brief

Project Overview

Make a motor controller and electronic steering configuration, a safety circuit that relays information from sensors, when certain sensors are showing certain figures shut down.

Motor Controller

Calibrate speed

Calculate turning

Calibrate power distribution from turning.

Safety Circuit

Finds flaws using gauges like volt meter, temperature gauge, incorrect figures

Safety Circuit shutdown

Once flaws are found shut down

Currently I've configured lower-level prototypes with less complex control systems but the key base is there for me to keep building off of.

No CAD designs have been made, nothing on the actual prototype but smaller prototypes related have been built.

Architecture has been defined by myself and I understand the power flow of how it will work / mechanical design.

Still to be done is CAD, implementation and configuration of components.

Primary Objectives

1. Motor Controller

Design a low-level motor controller capable of controlling two independent rear-wheel motors.

The controller should:

- Receive throttle input.

Calculate error and exact distribution between both motors. → smooth operation with control systems

- Receive steering input.

Calculate linear vectorisation with wheels / how much power each needs.

- Generate desired wheel speeds for the left and right motors.
- Distribute motor power between both wheels to produce steering electronically.

Send the calculated figures to both motors.

- Operate as a closed-loop control system using encoders for figures and readings.

2. Safety Circuit

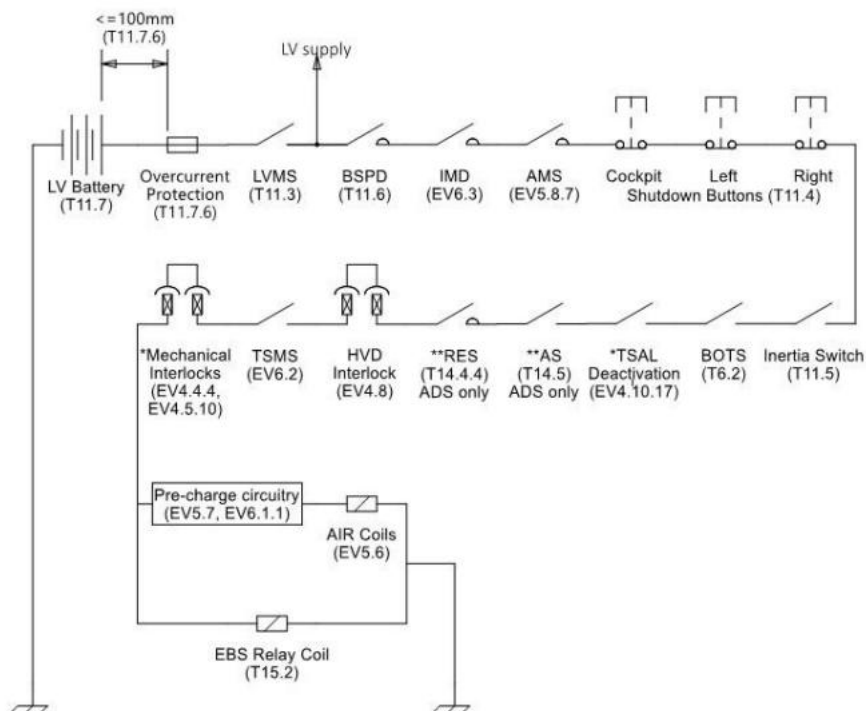
Design a hardware safety system that continuously monitors drivetrain health.

The system should monitor:

The stuff in the regulation

- Voltage
- Current
- Motor temperature
- Sensor failures
- Electrical isolation faults
- Emergency stop inputs

If dangerous operating conditions are detected, the safety system should trigger the shutdown circuit.

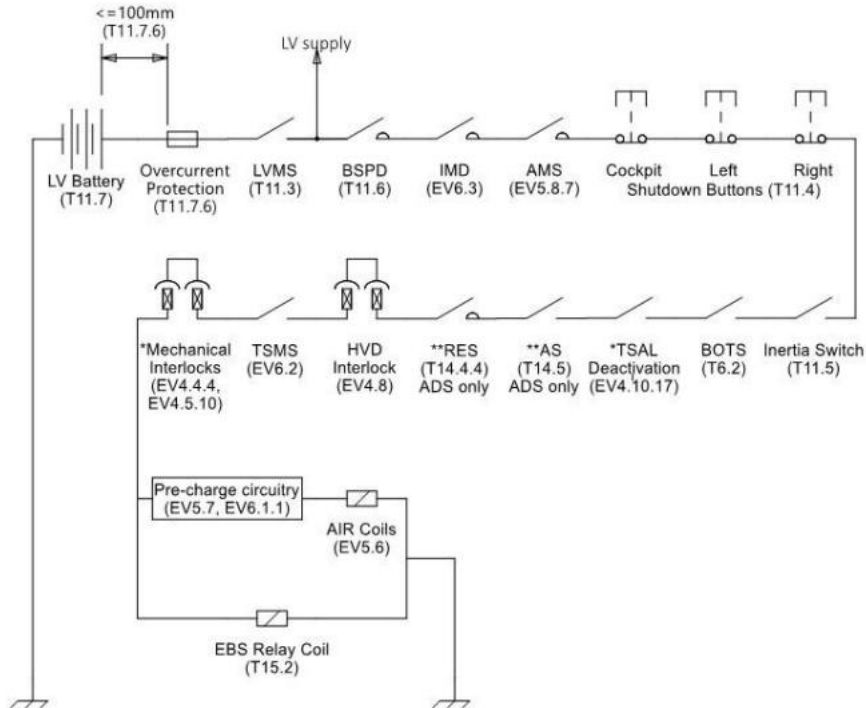


3. Shutdown Circuit

Design a shutdown system that complies with Formula Student regulations.

The shutdown system should:

- Immediately disable the drivetrain during hazardous conditions.
- Remove power from the motors.
- Latch faults requiring manual reset where required.
- Operate independently of programmable software where required by Formula Student rules.



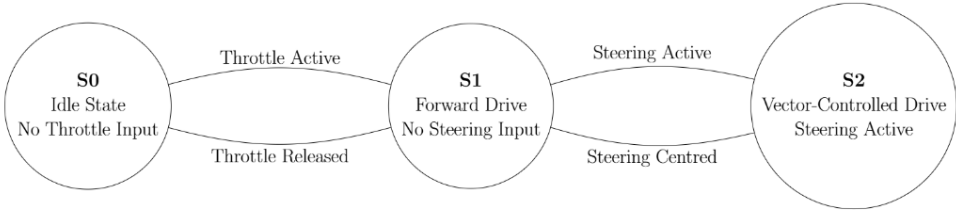
Functional Requirements

Two drive chains should act independently

- Drive two independent rear wheels.
- Use direct-drive electric motors.
- Use electronic steering through differential wheel speeds.
- Move a vehicle with a minimum mass of approximately 200 kg.
- Operate safely under normal and fault conditions.
- Be suitable for integration with a hydrogen fuel cell powertrain.

Control System Architecture

Low-Level Motor Controller State Diagram for Hydrogen Vehicle



State Descriptions

S0: Idle State
The drivetrain is powered but no throttle input is applied. The vehicle remains stationary.

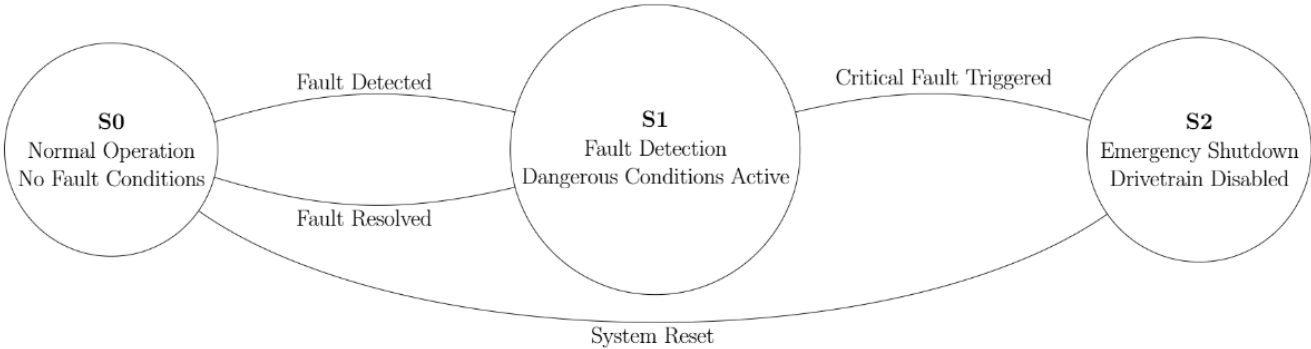
S1: Forward Drive
Throttle input is active and equal power is distributed to both motors, resulting in straight-line movement.

S2: Vector-Controlled Drive
Throttle and steering inputs are active. Motor power is dynamically adjusted between the left and right wheels using linear vectorisation to control turning.

Feedback from wheel encoders will be used to compare the desired speed with the measured speed, producing an error signal for closed-loop control.

Safety System Architecture

Drivetrain Safety and Shutdown State Diagram



State Descriptions

S0: Normal Operation
The drivetrain operates normally with no active fault conditions detected. Power is supplied to both motors and vehicle movement is enabled.

S1: Fault Detection
A dangerous operating condition has been detected within the drivetrain or control circuitry. The system monitors the fault condition and determines whether shutdown is required.

S2: Emergency Shutdown
A critical fault condition has triggered the shutdown circuit. Power to the drivetrain is immediately disconnected to ensure vehicle safety.

Current Understanding

The vehicle is expected to include:

- Hydrogen fuel cell power source.
- Battery system.
- Two independent rear electric motors.
- Rear-wheel drive.
- Direct-drive drivetrain.
- Electronic steering through differential motor control.
- Sensor feedback for safe operation.
- Formula Student compliant shutdown circuitry.

Current Development Roadmap

Software

- Differential steering calculations 😊
- Power distribution calculations 😊
- Bang-Bang controller 😊
- Proportional controller 😊
- PI controller 😊
- PID controller 😊
- Dual motor closed-loop controller → Add second motor

At University

- Wheel speed synchronisation → Both at the same time
- Current limiting
- Regenerative braking (if required) → sending both to max / Ben knows more about this

Safety

- Fault detection concept 😊
- Basic shutdown logic 😊
- Voltage monitoring
- Current monitoring
- Temperature monitoring
- Sensor validation
- Hardware safety circuit
- Formula Student shutdown implementation

Mechanical

Final drivetrain architecture

Motor selection

Battery selection

Fuel cell integration

CAD development

Packaging

Chassis integration

Formula Student Considerations

The drivetrain must comply with Formula Student EV regulations, including:

- Shutdown circuit requirements.
- Insulation Monitoring Device (IMD).
- Accumulator Isolation Relays (AIRs).
- Tractive System Master Switch (TSMS).
- Battery Management/Accumulator Management System (AMS).
- Brake System Plausibility Device (BSPD).
- Emergency shutdown buttons.
- Manual fault reset where required.
- Non-programmable shutdown logic for critical safety functions.

Unknowns

The following aspects are yet to be defined:

- Final vehicle layout.
- Motor specification.
- Battery specification.
- Fuel cell specification.
- Tractive system voltage.
- Motor controller hardware.
- Communication protocol (PWM, CAN, etc.).
- Steering mechanism implementation.
- Wheel encoder arrangement.
- Mechanical packaging.
- Suspension design.
- Final CAD models.
- Simulation workflow (SolidWorks Simulation, nTop, FEA).

Available Resources

Software:

- SolidWorks (University of Exeter)
- SolidWorks Simulation
- nTop (Formula Student Society)

Facilities:

- REEF
- University workshops
- Formula Student Society resources

Personal Project Goal

Develop a safe, reliable and well-documented electronic drivetrain that demonstrates:

- Closed-loop motor control.
- Electronic steering through differential wheel speed control.
- Fault detection and shutdown logic.
- Compliance with Formula Student safety requirements.

The long-term objective is to produce a drivetrain architecture that can be integrated into the team's hydrogen fuel cell vehicle and provide a foundation for future testing, CAD development and vehicle integration.